

④



Lec_4

Practice Questions on Time Complexity

④ $\Rightarrow$ Lec-3

Time complexity
 $\downarrow$

{ how many times loop is entered }
or
we Entered inside to loop

5.) Main()

```
{  
  while(n >= 1)  
  {  
    n = n - 2  
  }  
}
```

n = 12 10

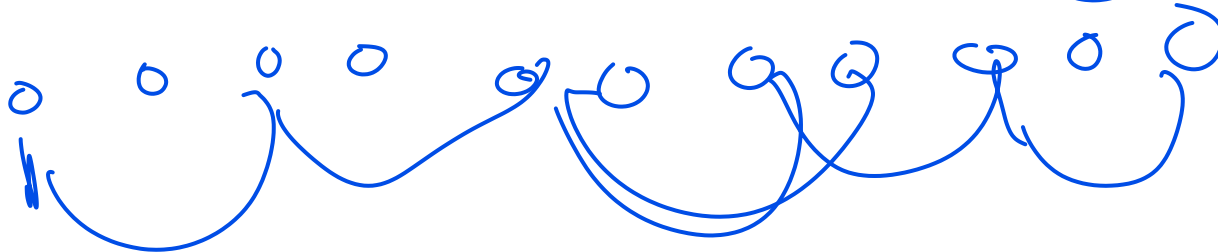
12 >= 1
10 >= 1
8 >= 1
6 >= 1
4 >= 1
2 >= 1
0 >= 1

n = 10
n = 8
n = 6
n = 4
n = 2
n = 0
n = -2

~~12~~ 10
n

n = 12 - 2
n = 10

6 7



12

$$n \rightarrow \underline{12}, \underline{10}, \underline{8}, \underline{6}, \underline{4}, \underline{2}, \underline{0}$$

$$\underline{n-2L=1}$$

$$n-2K \approx 0$$

$$\Rightarrow n=2K$$

$$\boxed{L=n/2}$$

$$n/2$$

$$O(n/2)$$

$$\Rightarrow \cancel{O(n)}$$

$$n=n-2$$

$$n/2$$

$$\Rightarrow n-3$$

$$0 \cup 0 \cup 0 \cup 0$$

$$n-100$$

$$\Rightarrow n/100$$

$$O(n/100)$$

$$O(n/3)$$

$$O(n)$$

$$n=n-10$$

$$n/10$$

```
6.) Main()
{
  while(n >= 1)
  {
    n = n - 5
  }
}
```

20, 15, 10, 5, 0

$$20/5 = 4$$

$$O(n/5) \Rightarrow \underline{O(n)}$$

$$n/5$$

$$n \cdot 5(1) = 0$$
$$1 \cdot n/5$$

$$\underline{n - a} \Rightarrow$$

$a = \text{constant}$

$$n+1 \Rightarrow O(n)$$

$$n+20 \Rightarrow O(n)$$

$$n/1000 \Rightarrow O(n)$$

$$O(1) + O(1) + O(1) + O(1) = O(1)$$

while($n \geq 1$)

$$n = n - a$$

~~total~~

$$\Rightarrow O(n/a)$$

$$= O(n)$$

1000

$n \geq 1$

~~$n = 1000$~~

~~900~~

~~700~~

~~750~~

~~$n = 750$~~

$\rightarrow n/9$

$n - a$

~~$O(n/250)$~~

~~$O(n)$~~

$n = n - 200$
ur $\rightarrow$ pre

$n = n - 300 + 10$

$n = n - 250$

7.) Main()

{

while($n \geq 1$)

{

$n = n - 200$

$n = n - 100$

$n = n + 50$

}

}

$n = n - 250$

$n = n - 250$

1 ✓
2 ✓
3 ✓

$$\begin{aligned}n &= n - 100 \\n &= n - 200 \\n &= n + 1000 \\n &= n + 50\end{aligned}$$

$$\textcircled{\textcircled{n = n + 700}}$$

8.) Main()

{ i=1

while(i<=n)

{

i=i+3

i=i+5

i=i+9

}

}

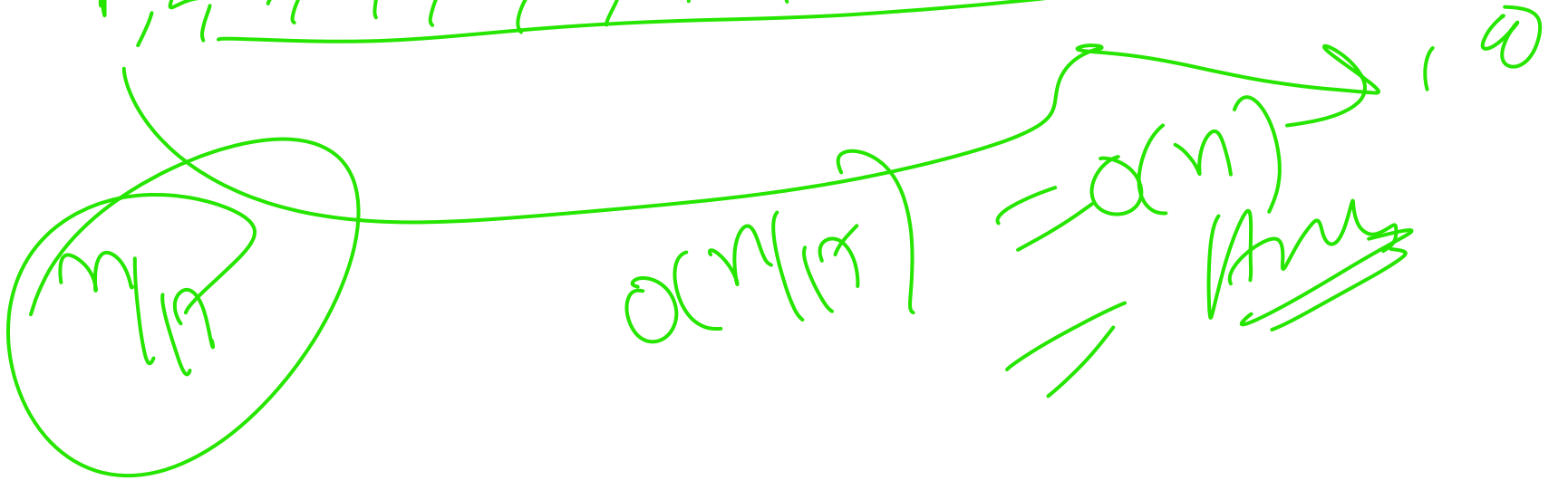
~~i=1~~ ~~4~~ ~~9~~ 16 $\rightarrow$ 17

~~i=10~~ ~~21~~ ~~26~~ 35

~~i=30~~ ~~48~~ 52

0 0 0 0 $\rightarrow$ $O(N) = O(N)$ 17

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17



$l < \underline{n^2}$

```
9.) Main()  
{ i=1  
  while(i<=n^2)  
  {  
    i=i+3  
    i=i+5  
    i=i+9  
  }  
}
```

ny l=1, .

$l = i + 17$

$\frac{n^2}{17}$ X

$\frac{n^2}{1}$

$\frac{n^4}{15}$

$\Rightarrow O(n^2)$

$\Rightarrow \underline{O(n^4)}$